

Replication: Early and Late Human Capital Investments, Borrowing Constraints, and the Family

Caucutt & Lochner (2020, *Journal of Political Economy*)

Replication by Jared Miller

Abstract

This document reports a full replication of the quantitative results in Caucutt and Lochner (2020), covering motivating data work (Table 1), the calibrated baseline (Tables 2-7) and the policy counterfactuals (Tables 8–15), followed by an original counterfactual crossing investment subsidies with borrowing-constraint regimes. When elements of the replication are inconsistent with the published results, tables are replicated twice, one with the initial Endogenous grid method solver used, and the other with a Nelder-Mead solver translated from the paper’s Fortran code into Julia. The majority tables replicate to within rounding. Where they do not—the short run of Table 13 and the welfare-maximizing reform of Table 15—I document and discuss the discrepancies.

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1 Introduction

Caucutt and Lochner (2020) study what drives the parental investment and educational attainment gap across the income distribution. In particular, they examine (1) the role of wage uncertainty, (2) the transmission of child ability, and (3) borrowing constraints. The paper uses NLSY data to identify a dynastic parental investment model, matching moments across the education distribution, the wealth distribution, and among others. Breaking childhood into two segments, the paper finds roles for all three of these mechanisms, and in particular identifies **borrowing constraints** as a **major player in the investment gap**.

2 Model Overview

Agents live for six periods, each corresponding to twelve years; for the first two periods, children are dependent on their parents for consumption and investment choices. Periods 1 and 2 correspond to early and late childhood, periods 3 and 4 to young and old parenthood, and periods 5 and 6 to post-parenthood and retirement.

During the parenting stages, agents make early and late investment decisions i'_1, i'_2 that feed into their child's future human capital $h' = \theta' f(i'_1, i'_2)$. When the parent and child separate, parents also make a parental transfer decision a'_3 . Each asset decision comes with a **borrowing constraint** $a_j \geq -L_{j-1}(h)$ that is a **fraction of the agent's natural borrowing limit**. Finally, ability shocks ($\theta'_H > \theta'_L$) depend on both the parent's ability θ and human capital h , and at each stage a worker faces a wage shock $W_j(h) = \Gamma_j(h + \epsilon_j)$, where $\log(\epsilon_j) \sim N(m, s^2)$.

3 Initial Code Divergence

The steady-state distribution results from the paper's code and from my original submission were not close. The fundamental ways the two codebases misaligned at the start were:

1. **State-space consolidation.** The paper consolidates $Ra + \varepsilon$ into a single state z . Advantage: instead of discretizing the wage shock ε , one makes an investment choice $a \rightarrow Ra$, and any movement up the discrete z grid is interpreted as the realization of the ε shock. I switched from discrete ε grids to this method, which delivers more grid points when taking expectations, and greater computational efficiency.
2. **Asset grid.** My code used one asset grid for all agents, with complicated feasibility checks ensuring no agent carried debt above their limit. The paper's code uses a unique asset grid for each (h, t) tuple, which is far better. The paper also increases the upper bound as agents age, which matters for wealth accumulation.
3. **Model error and solution method.** My c_3 guess was inconsistent with my V_3 guess, and the old-parent intratemporal first-order condition does not always hold, which I had originally

treated as if it did. The paper uses fitted VFI nested inside discrete investment choices, while my code uses EGM nested inside discrete investment choices.

4 Solution-Method Details and Fixes

4.1 Post-Parent's Problem

$$V_5(h, z_5) = \max_{a_6} \frac{(z_5 + h - a_6)^{1-\sigma}}{1-\sigma} + \beta(Ra_6)^{1-\sigma}$$

This admits the simple analytical solution $a_6 = \frac{z_5+h}{1+(\beta R)^{-1/\sigma}}$. Since the old-parent's problem requires it, we take the expectation over ε given a_5 (where $z' = Ra_5 + \varepsilon_5$):

$$E[V_5(h, z_5) \mid a_5] = \sum_{z_{5j}=Ra_5}^{z_{5j+1}} \int_{z_{5j}}^{z_{5j+1}} \left[V_5(z_{5j}, h) + \frac{V_5(z_{5j+1}, h) - V_5(z_{5j}, h)}{z_{5j+1} - z_{5j}} (z' - z_{5j}) \right] f(\varepsilon_5) d\varepsilon.$$

My original code computed this in discrete jumps with a fixed V_5 . The paper's code linearly interpolates, in which case the expression above has a closed-form solution.¹ The resulting values are identical.

4.2 The Log-Normal Expectation Trick

Pulling constants out of the expectation above, letting $\Psi = \frac{V_5(z_{5j+1}, h) - V_5(z_{5j}, h)}{z_{5j+1} - z_{5j}}$, and recalling that $z' = \underbrace{Ra_5}_{\text{fixed}} + \varepsilon_5$:

$$\begin{aligned} \Psi \int_{z_{5j}}^{z_{5j+1}} \varepsilon_5 f_{LN}(\varepsilon_5; m_5, v_5) d\varepsilon_5 &= \Psi \int_{z_{5j}}^{z_{5j+1}} f_{LN}(\varepsilon_5; m_5 + v_5, v_5) d\varepsilon_5 \\ &= \Psi \exp\left(m_5 - \frac{v_5}{2}\right) \left(F_{LN}(z_{5j+1} - Ra_5; m_5 + v_5, v_5) - F_{LN}(z_{5j} - Ra_5; m_5 + v_5, v_5) \right). \end{aligned}$$

With this convenient algebraic property, discretization error is mitigated when taking expectations.

4.3 Final Fix: The Old-Parent's Problem

$$\begin{aligned} V_4(i_1, h, z_4, \theta') &= \max \left\{ u(c_4) + \rho u(c'_2) + \beta \left(E_{z_5}[V_5(h, z_5) \mid a_5] + \rho E_{z_3, \theta''}[V_3(h', z'_3, \theta'') \mid a'_3, \theta', h'] \right) \right\} \\ \text{subject to } &c_4 + c'_2 + a'_3 + a_5 = z_4 + \Gamma_2 h + W_2 - \Phi(i'_2), \\ &a'_3 \geq \max(-L_2(h'), W_2 - \Phi(i'_2) - c'_2), \quad a_5 \geq -L_4(h), \quad h' = \theta' f(\tilde{i}'_1, p_2 + i_2). \end{aligned}$$

¹This is a fix I apply to all other expectations as well. Mine was not technically incorrect, but it could have been much more precise.

The optimality conditions are

$$\underbrace{c_4^{-\sigma} \geq \rho c_2'^{-\sigma}}_{\text{intratemporal}}, \quad \underbrace{u'(c_4) \geq \beta R E_{z_5}[u'(c_5)]}_{\text{old-parent saving}}, \quad \underbrace{u'(c_2') \geq \beta R E_{z_3', \theta''}[u'(c_3')]}_{\text{transfer to child}}.$$

The intratemporal condition holds with equality only when the non-negative transfer constraint is slack, and the transfer Euler holds with equality if and only if $a_3' \geq -L_2(h')$ is slack.²

4.4 An Inconsistency Between the Paper’s Code and the JPE Write-Up

One **inconsistency** found while replicating is the treatment of the unobserved costs of schooling, ζ , in the code versus the published paper. Unobserved costs are split into a high-school component ζ_1 and a college component ζ_2 . In the JPE, these are framed as additive costs across the four educational attainment choices i_2' : $\zeta(i_2') = 0$ for HS dropouts, $2\zeta_1$ for HS graduates, $2\zeta_1 + 2\zeta_2$ for some college, and $2\zeta_1 + 5\zeta_2$ for college graduates. In the Fortran code, the unobserved costs are $[0, 2\zeta_1, 2\zeta_2, 3\zeta_2]$: rather than being added, the components are overwritten. The model consistently uses the latter throughout, but the interpretation of the calibrated parameters differs.

4.5 Solver Comparison: EGM versus Fitted VFI

Using a method that avoids numerical optimization and exploits the optimality conditions directly yields a significant speedup at a convergence tolerance of 10^{-8} : the EGM code solves the baseline model in ≈ 93 seconds, while the fitted VFI translation takes ≈ 707 seconds—a **7.5 $\times$ speedup**.

Comparing the two fixed points, $\|V_{3,\text{repl}} - V_{3,\text{paper}}\|_\infty = 2.2 \times 10^{-7}$, or -0.86% . The worst cell is the wealthiest, highest-human-capital parent, where all consumption and savings policies are interior and optimization packages mechanically beat EGM. For the old-parent’s problem, $\|V_{4,\text{repl}} - V_{4,\text{paper}}\|_\infty = 9.5 \times 10^{-7}$, or -3.8% . The story is the same: the worst cell is the wealthiest z_4 cell, where all policies are interior, and discretization error is largest because grid points are sparsest at high z .

²My original code made mistakes in both of these cases.

5 Motivating Data Analysis (Table 1)

	Earned Income								Earned "Full" Income							
	Sample Size		Early Income		Later Income		Eq. Effects (<i>p</i>)		Sample Size		Early Income		Later Income		Eq. Effects (<i>p</i>)	
	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
A. Controls Only for Maternal Education																
HS dropout (ages 21–24)	2,273	2,280	−.026** (.004)	−.026** (.004)	−.009 (.006)	−.009 (.006)	.063	.066	1,894	2,078	−.023** (.005)	−.023** (.004)	−.011 (.006)	−.011 (.006)	.205	.165
Attended college (ages 24–27)	1,586	1,675	.046** (.007)	.046** (.007)	.015 (.009)	.015 (.009)	.037	.031	1,336	1,531	.045** (.008)	.044** (.007)	.014 (.010)	.012 (.009)	.044	.023
Graduated college (ages 24–27)	1,586	1,675	.028** (.006)	.027** (.005)	.024** (.007)	.024** (.007)	.765	.823	1,336	1,531	.033** (.009)	.026** (.005)	.025** (.008)	.026** (.007)	.500	.966
B. Control for Maternal Education and Child/Family Background																
HS dropout (ages 21–24)	2,190	2,196	−.023** (.005)	−.024** (.005)	−.010 (.006)	−.011 (.006)	.173	.183	1,835	2,012	−.022** (.005)	−.021** (.005)	−.012 (.006)	−.012** (.006)	.295	.337
Attended college (ages 24–27)	1,524	1,610	.040** (.008)	.040** (.008)	.011 (.009)	.011 (.009)	.051	.045	1,291	1,479	.042** (.008)	.039** (.008)	.008 (.010)	.006 (.009)	.035	.028
Graduated college (ages 24–27)	1,524	1,610	.025** (.006)	.023** (.006)	.021** (.007)	.021** (.007)	.778	.854	1,291	1,479	.032** (.006)	.024** (.006)	.021** (.008)	.022** (.007)	.361	.851

Table 1: Effects of Early and Late Income on Child Educational Attainment. Standard errors are in parentheses. P = Original Paper, R = Replication.

6 Baseline Results (Tables 2–7)

For all results below, I strictly follow the paper’s methodology: baseline results iterate to a stationary distribution, and where simulated data sets are needed, I use identical methodology (number of observations, seed files, etc.). Each baseline table is reported twice: once with my EGM solver and once with the Nelder–Mead (NM) Julian translation of the paper’s own solver. In particular, the Nelder–Mead solve was changed to a higher tolerance, and added corner constraints checks for all cells. This should only lead to improvement in the solver finding the true optimum.

6.1 Table 2: Calibrated Education Distribution

Education	Paper Model	Replication (EGM)
HS grad or more	.83	.84
Some college+	.44	.44
College graduate	.21	.22

Table 2: Calibrated education distribution, EGM solver (paper Table 2).

Education	Paper Model	Replication (NM)
HS grad or more	.83	.83
Some college+	.44	.44
College graduate	.21	.21

Table 3: Calibrated education distribution, NM translation solver (paper Table 2).

6.2 Table 3: Investment Scores by Parental Education

Investments in dollars are mapped to factor scores by the function $\Phi(i_1) = -0.21 + 0.00085 i_1 - 0.0000001 i_1^2$, calibrated from NLSY moments on early investment factor scores.

Parent Educ.	Score		HS+		SC+		CG	
	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl
HS dropout	-.49	-.52	.64	.62	.20	.19	.08	.07
HS graduate	-.40	-.37	.81	.84	.27	.27	.08	.09
Some college	.11	.17	.90	.90	.57	.57	.15	.15
College grad	.67	.67	.94	.94	.82	.82	.63	.64

Table 4: Investment scores by parental education, EGM (paper Table 3).

Parent Educ.	Score		HS+		SC+		CG	
	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl
HS dropout	-.49	-.48	.64	.64	.20	.20	.08	.08
HS graduate	-.40	-.38	.81	.81	.27	.27	.08	.08
Some college	.11	.16	.90	.90	.57	.57	.15	.15
College grad	.67	.67	.94	.94	.82	.82	.63	.63

Table 5: Investment scores by parental education, NM (paper Table 3).

6.3 Table 4: Investment Scores by Parental Income

Inc. Qtile		Score		HS+	
Early	Late	Paper	Repl	Paper	Repl
1	Any	-.56	-.56	.73	.74
2	Any	-.43	-.41	.81	.82
3, 4	Any	.36	.57	.89	.90
Any	1	-.36	-.34	.71	.72
Any	2	-.24	-.18	.81	.82
Any	3, 4	.16	.48	.90	.91
1	1	-.52	-.52	.66	.67
1	2	-.60	-.60	.73	.74
2	2	-.44	-.43	.80	.80
2	3, 4	-.39	-.36	.88	.89
3, 4	3, 4	.49	.67	.92	.93

Table 6: Investment scores by parental income, EGM (paper Table 4).

Inc. Qtile		Score		HS+	
Early	Late	Paper	Repl	Paper	Repl
1	Any	−.56	−.56	.73	.73
2	Any	−.43	−.42	.81	.81
3, 4	Any	.36	.56	.89	.89
Any	1	−.36	−.34	.71	.71
Any	2	−.24	−.18	.81	.81
Any	3, 4	.16	.47	.90	.90
1	1	−.52	−.52	.66	.66
1	2	−.60	−.59	.73	.73
2	2	−.44	−.43	.80	.80
2	3, 4	−.39	−.37	.88	.88
3, 4	3, 4	.49	.66	.92	.92

Table 7: Investment scores by parental income, NM (paper Table 4).

Discussion of Table 4. Investigating the highlighted differences, the results are consistent with a discrete investment jump at the highest resource points, and thus are more sensitive to tolerance choices and discretization error. The Nelder-Mead result comes from a tightening of the objective tolerance ($tol = 1.0 \times 10^{-14}$). With these tighter tolerances, the replication solvers agree with each other.

6.4 Table 6: Baseline Investment Amounts by Parental Education

Parent Educ.	i_1		i_2		$i_2 - S_2$		$i_2 + \zeta - S_2$	
	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl
All	1,888	1,898	8,744	8,861	4,757	4,822	5,629	5,702
HS dropout	770	712	4,351	4,047	2,262	2,093	2,671	2,478
HS graduate	907	932	5,217	5,439	2,691	2,810	3,212	3,345
Some college	1,857	1,870	8,739	8,832	4,713	4,767	5,716	5,778
Coll. grad.	4,600	4,588	18,687	18,774	10,563	10,615	12,304	12,359

Table 8: Baseline investment amounts by parental education, EGM (paper Table 6).

Parent Educ.	i_1		i_2		$i_2 - S_2$		$i_2 + \zeta - S_2$	
	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl
All	1,888	1,888	8,744	8,781	4,757	4,779	5,629	5,654
HS dropout	770	763	4,351	4,343	2,262	2,257	2,671	2,665
HS graduate	907	910	5,217	5,250	2,691	2,711	3,212	3,235
Some college	1,857	1,865	8,739	8,822	4,713	4,762	5,716	5,771
Coll. grad.	4,600	4,594	18,687	18,721	10,563	10,584	12,304	12,324

Table 9: Baseline investment amounts by parental education, NM (paper Table 6).

The investment-amount tables are noisier than the others. Near-indifference between adjacent discrete choices, combined with solution-method differences, is the likely culprit. As per the NM

translated version, Table 6 is much tighter.

6.5 Table 7: Fraction of Parents Constrained

Parent Educ.	Young		Old		Transfer	
	Paper	Repl	Paper	Repl	Paper	Repl
All	.12	.11	.14	.29	.00	.00
HS dropout	.13	.13	.06	.08	.01	.01
HS graduate	.20	.19	.17	.24	.00	.00
Some college	.06	.06	.17	.41	.00	.00
Coll. grad.	.01	.01	.14	.42	.00	.00

Table 10: Fraction of parents constrained, EGM (paper Table 7).

Parent Educ.	Young		Old		Transfer	
	Paper	Repl	Paper	Repl	Paper	Repl
All	.12	.12	.14	.29	.00	.00
HS dropout	.13	.13	.06	.08	.01	.01
HS graduate	.20	.20	.17	.25	.00	.00
Some college	.06	.06	.17	.41	.00	.00
Coll. grad.	.01	.01	.14	.42	.00	.00

Table 11: Fraction of parents constrained, NM (paper Table 7).

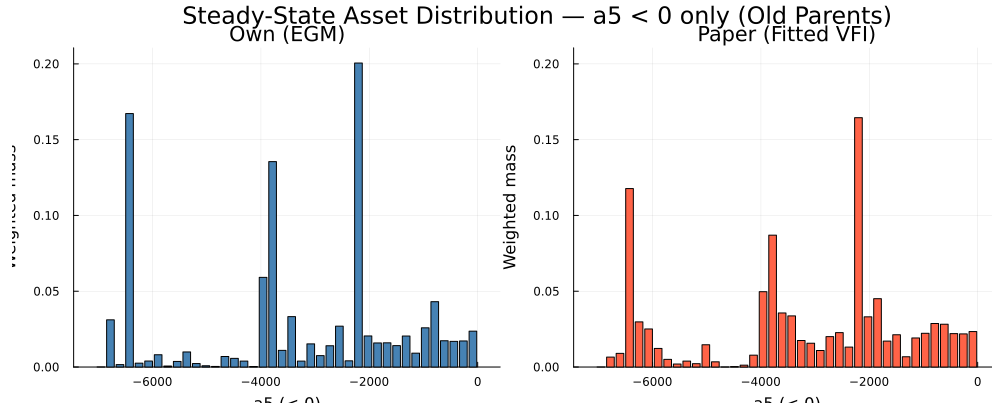


Figure 1: Distribution of old-parent savings a_5 in the negative region. The spikes where EGM clamps to the borrowing limit are dampened in the paper’s version. Note the paper version use the Fortran code’s tolerances from the replication files.

The old-parent constrained fraction is the one baseline moment with real divergence from the published table. Figure 1 shows why: EGM clamps savings exactly to the limit, while a derivative-free optimizer approaches the floor over a very flat objective without exactly hitting it, so whether

a cell is counted as “constrained” depends on the corner tolerance. Re-running both solvers with tight tolerances and explicit manual corner checks leads to agreement between the two-solvers.

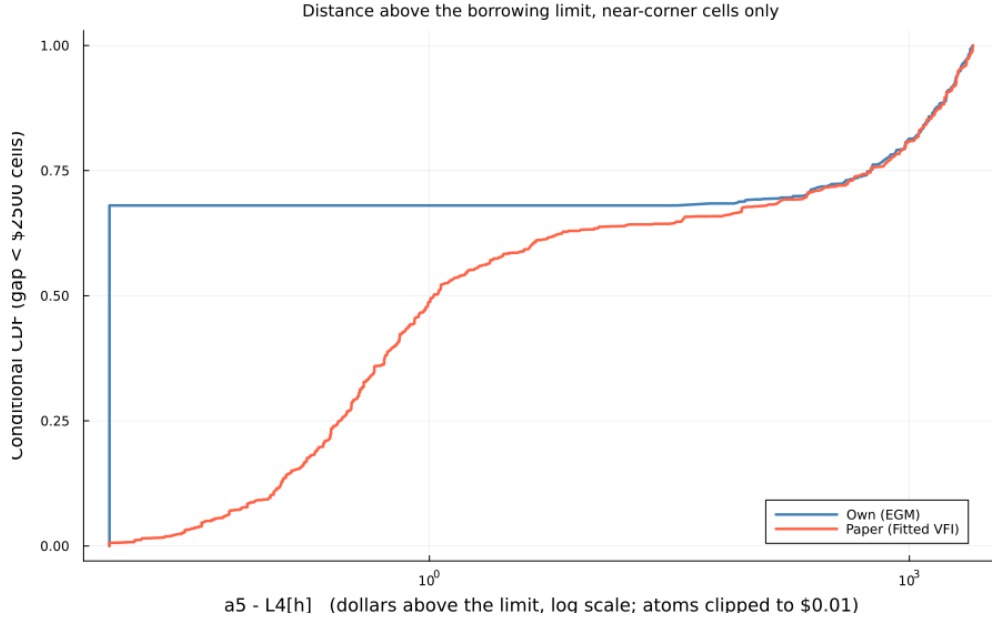


Figure 2: CDF of post-parent a_5 asset distribution. Same story is seen here, CDF is lower in the Fitted VFI case when agents are near their borrowing constraints, and nearly identical elsewhere

Gap Bin	Own (wt mass)	Paper (wt mass)
[0.0, 1.0)	0.290623	0.210147
[1.0, 10.0)	0.000000	0.062172
[10.0, 100.0)	0.004674	0.019727
[100.0, 1000.0)	0.052201	0.056266

Table 12: Distribution of $(a_5 - L_4)$ Gap using own EGM versus NM with Paper’s tolerances

Discussion of Table 7. and Related Figures The figures show what is going on. The EGM solution puts about .29 of old-parent mass exactly at the borrowing limit, while the NM solution has virtually little mass exactly at the limit, with that same mass instead sitting between roughly \$0.10 and \$300 above it. Counting everything within \$100 of the limit, the two solutions agree almost exactly (.295 vs .292). The reason NM never lands on the limit is mechanical: Nelder-Mead approaches the limits, but to reach a manual corner check, the optimizer, which is bounded by the limit, must get to within \$0.0025 of the borrowing limit. This is due to the exponentially spaced grid points and since z_5 is cheap for the post-parent’s problem, there are 360 grid points in total. Thus, the reported fraction of agents constrained is sensitive to tolerance choices. Only adding manual corner checks leads to too many reported constrained, so the combination of finer tolerances and manual corner checks gets the Nelder-Mead solver closer to the true optimum.

7 Policy Counterfactuals (Tables 8–15)

7.1 Methodology

All counterfactuals below follow the stationary-distribution logic of the paper’s `policy.f90`. When there is notable deviation from the published tables, we re-run the policy counterfactual using the Nelder-Mead translated solver. For a permanent policy, decision rules are time-invariant from the moment the policy lands, so the two horizons differ only through the distribution being aggregated over. When panel data is required, the seeds from the replication file are used:

- **Short run:** The new decision rules are applied to the *baseline* stationary distribution—the population as it stands the day the policy is implemented.
- **Long run:** the density iteration is run to convergence under the new rules, and moments are re-aggregated on the *new* stationary distribution.

One-time (non-permanent) policies are handled with the limited value-iteration passes the Fortran code uses, i.e., your child’s V_3 is identical to the baseline case, and we resolve the Value Functions using the one-time policy regime.

7.2 Table 8: One-Time Income/Wealth Transfers ($\Delta\%$)

Policy	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		CG		$\bar{W}_3$	
	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl
\$10k unanticip.	.0	.0	1.4	1.5	1.2	1.4	3.0	3.2	.0	.0	.2	.2
\$10k anticip.	8.0	8.4	6.2	6.7	.5	.7	7.2	7.3	8.5	9.4	1.3	1.3
\$10k/ R young	9.0	9.1	7.0	6.9	.9	1.0	8.0	7.8	9.6	9.4	1.4	1.4

Table 13: Short-run effects of one-time income/wealth transfers, % change (paper Table 8).

7.3 Table 9: Decomposition of Investment Gaps

	$\bar{i}_1$		$\bar{i}_2$		SC+	
	Paper	Repl	Paper	Repl	Paper	Repl
<i>Investment gaps (\$)</i>						
Baseline (uncond.)	3,057	3,067	7,743	7,796	.38	.38
Cond. parent ability	2,940	2,950	6,938	6,984	.34	.34
Cond. child ability	2,615	2,625	5,924	5,977	.28	.28
<i>Relax all limits: % Δ from baseline</i>						
Unconditional	+16.3	+16.4	+5.6	+6.3	−14.1	−15.2
Cond. child ability	−18.9	−18.9	−51.5	−50.4	−67.6	−68.7

Table 14: Decomposition of investment gaps between parental income quartiles 1 and 4 (paper Table 9).

7.4 Table 10: Intergenerational Ability and Investment Transmission

	(1)		(2)		(3)		(4)	
	Paper	Repl	Paper	Repl	Paper	Repl	Paper	Repl
<i>Avg. gap (\$): coll. grad. – HS dropout</i>								
i_1	3,829	3,876	3,468	3,295	2,525	2,439	5,385	5,410
i_2	14,336	14,727	13,080	12,739	9,092	8,864	21,680	21,919
<i>Intergenerational correlation</i>								
In θ	.31	.31	.31	.31	.00	.00	1.00	1.00
In i_2	.52	.53	.46	.46	.29	.28	.85	.86
In h	.50	.50	.46	.46	.28	.27	.87	.87
In lifetime earnings	.29	.30	.26	.26	.19	.19	.44	.44

Table 15: Intergenerational ability and investment transmission (paper Table 10). Columns: (1) baseline; (2) no $h \rightarrow \theta'$ channel; (3) no parent–child ability correlation; (4) perfect correlation.

7.5 Table 11: Young-Parent Borrowing Limit +\$2,500 ($\Delta\%$)

Parent Educ.	Short Run										Long Run									
	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$		$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$	
	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
All	2.6	2.7	1.9	1.8	.5	.1	4.8	4.8	.4	.4	−6	−5	−6	−7	.1	−.4	1.5	1.5	−.1	−.1
HS dropout	3.2	3.3	3.2	2.6	3.9	2.0	5.4	5.3	.3	.3	1.4	−.0	1.7	−.1	3.1	1.0	2.3	−.3	.1	−.1
HS graduate	5.8	6.4	3.2	3.1	.1	−.3	8.2	8.6	.5	.5	−.8	1.1	−1.1	−.3	−.6	−.8	.5	2.1	−.1	.0
Some college	4.7	4.5	3.7	3.6	−.4	−.1	7.9	7.5	.8	.8	−.8	−.7	−.5	−.3	−.8	−.6	3.1	2.8	−.1	−.0
Coll. grad.	.5	.5	.1	.1	.0	.0	.4	.4	.1	.1	.5	.5	.1	.1	.0	.0	.4	.4	.1	.1

Table 16: Effects of increasing young parents' borrowing limit by \$2,500, % change (paper Table 11). P = Paper, R = Replication.

7.6 Table 12: Old-Parent Borrowing Limit +\$2,500 ($\Delta\%$)

Parent Educ.	Short Run										Long Run									
	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$		$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$	
	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
All	10.9	11.2	9.6	9.4	2.2	2.2	5.0	4.6	1.8	1.8	11.0	11.0	9.8	9.5	2.5	2.4	5.3	4.9	1.8	1.8
HS dropout	13.2	13.0	9.6	11.5	4.1	6.9	4.2	4.0	1.0	1.0	23.0	20.2	16.2	15.9	4.7	6.6	9.1	7.0	1.7	1.5
HS graduate	13.4	13.7	10.0	9.0	3.2	2.1	8.4	7.1	1.2	1.2	8.4	9.6	6.5	6.2	2.2	1.5	3.4	3.5	.5	.6
Some college	18.9	19.3	15.8	15.7	2.0	1.7	9.1	8.7	3.0	3.0	5.2	5.6	5.5	5.7	1.5	1.1	1.0	.4	.9	.9
Coll. grad.	6.3	6.6	6.1	6.1	−.3	.4	.1	.1	1.9	2.0	.0	.2	.7	.6	−.3	.1	−1.0	−.0	.1	.1

Table 17: Effects of increasing old parents' borrowing limit by \$2,500, % change (paper Table 12). P = Paper, R = Replication.

Discussion of Tables 11 and 12. Large majority of moments are hit. Likely the differences are due to the discretization error and discre-investment flips previously discussed. Those changes are more notable when conditioning on less-dense sub groups, particularly HS dropouts.

7.7 Table 13: Fully Relaxing All Borrowing Limits ($\Delta\%$)

Parent Educ.	Short Run										Long Run									
	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$		$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$	
	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
All	72.5	123.2	63.2	96.0	12.5	9.8	31.0	51.7	11.7	18.8	111.1	110.0	89.6	88.0	14.4	10.5	48.8	48.3	17.7	17.5
HS dropout	151.0	206.3	112.9	147.9	31.1	20.1	85.5	118.1	11.8	14.8	54.1	147.9	46.7	100.2	17.7	14.3	24.7	64.4	4.4	10.1
HS graduate	161.4	281.6	116.4	187.7	14.4	14.2	74.8	133.5	14.0	24.2	183.5	173.4	126.3	115.7	13.8	9.2	71.7	69.3	13.4	12.8
Some college	102.9	189.8	90.9	139.7	7.3	6.3	24.0	38.1	16.3	27.1	40.9	39.6	44.1	43.1	4.6	1.0	.8	.0	7.5	7.0
Coll. grad.	16.8	26.0	13.0	17.4	4.7	1.4	-.4	.6	4.5	6.5	12.1	11.8	9.7	9.6	4.5	2.6	-1.5	-1.6	3.3	3.2

Table 18: Fully relaxing all borrowing limits ($\gamma = 0.99$), % change, EGM with current-generation short-run rules (paper Table 13). P = Paper, R = Replication.

Parent Educ.	Short Run										Long Run									
	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$		$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		$\bar{W}_3$	
	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM
All	123.2	122.3	96.0	95.7	9.8	12.2	51.7	51.5	18.8	18.7	110.0	110.6	88.0	88.4	10.5	11.1	48.3	48.2	17.5	17.6
HS dropout	206.3	198.5	147.9	148.4	20.1	35.8	118.1	115.1	14.8	14.5	147.9	103.3	100.2	81.4	14.3	22.6	64.4	49.8	10.1	7.7
HS graduate	281.6	277.7	187.7	185.6	14.2	13.0	133.5	133.7	24.2	24.0	173.4	177.9	115.7	118.8	9.2	8.3	69.3	71.3	12.8	13.2
Some college	189.8	189.4	139.7	140.4	6.3	9.2	38.1	38.1	27.1	27.2	39.6	39.5	43.1	42.2	1.0	.8	.0	-.1	7.0	6.8
Coll. grad.	26.0	26.8	17.4	17.5	1.4	2.1	.6	.6	6.5	6.6	11.8	12.3	9.6	9.8	2.6	3.1	-1.6	-1.7	3.2	3.3

Table 19: Table 13 solver comparison. Short run: EGM current-generation rules vs. NM fully-converged permanent rules. Long run: both solvers on the converged $\gamma = 0.99$ steady state. EGM = endogenous grid method, NM = Nelder–Mead.

Discussion of Table 13. Magnitudes are all consistent with the published parameters, but percent changes are off in some areas in both cases for the solvers. The results are more consistent across the two replication model solvers than they are with the published results, with the only noticeable changes are conditioning on HS dropouts. In the long-run, HS dropouts shift from 16% $\rightarrow$ 7.5%, and thus those results are sensitive to discrete investment flips.

7.8 Table 14: Early- and Late-Investment Subsidies ($\Delta\%$)

Policy	Short Run												Long Run											
	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		CG		$\bar{W}_3$		$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		CG		$\bar{W}_3$	
	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
$s_1=.10$ (early)	63.6	64.2	22.5	22.8	.8	1.8	13.5	13.5	43.2	43.1	6.5	6.6	76.2	76.4	30.1	30.0	1.7	2.2	18.8	18.8	56.4	56.1	8.4	8.5
$s_2=.026$ (early)	12.9	13.2	25.9	25.5	15.9	14.7	17.7	17.5	39.3	39.1	3.6	3.6	17.2	17.4	29.6	29.1	16.2	15.0	20.6	20.3	45.5	45.2	4.4	4.4
$s_2=.026$ (late)	.0	.0	15.4	15.2	15.9	14.7	15.4	15.3	15.1	15.5	1.6	1.6	17.2	17.4	29.6	29.1	16.2	15.0	20.6	20.3	45.5	45.2	4.4	4.4

Table 20: Effects of early- and late-investment subsidies, % change (paper Table 14). P = Paper, R = Replication.

7.9 Table 15: Welfare-Maximizing Budget-Neutral Reform ($\Delta\%$)

	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		CG		$\bar{W}_3$	
	P	R	P	R	P	R	P	R	P	R	P	R
Change in averages / prob.	330.2	223.3	-47.0	-54.8	-15.2	-17.4	-19.5	-28.8	-90.5	-100.0	11.1	4.5
Change in gaps by educ.	192.5	66.6	-51.6	-62.0	20.8	12.8	-4.1	.9	-81.5	-100.0	16.9	-15.3

Table 21: Short-run effects of the welfare-maximizing budget-neutral reform ($s_1^* = 0.43$, $s_2^* = -0.356$), EGM (paper Table 15). P = Paper, R = Replication.

	$\bar{i}_1$		$\bar{i}_2$		HS+		SC+		CG		$\bar{W}_3$	
	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM	EGM	NM
Change in averages / prob.	223.3	223.6	-54.8	-54.8	-17.4	-17.4	-28.8	-28.7	-100.0	-100.0	4.5	4.5
Change in gaps by educ.	66.6	66.3	-62.0	-62.1	12.8	12.9	.9	.6	-100.0	-100.0	-15.3	-15.4

Table 22: Table 15 solver comparison at $s_1^* = 0.43$, $s_2^* = -0.356$: the two solver families agree to a decimal on every entry.

Discussion of Table 15. Again the percent changes are off as compared to the published version, but in this case the two independent Julia solvers nearly exactly agree.

7.10 Summary

The replication for Tables 8-12, and Table 14 are very close to the paper’s published results. Table 13 and Table 15 have various moments that were not replicated as per the published results, yet the Nelder-Mead and EGM Julia version exactly agreed with each other, and thus in all likelihood these results are perhaps sensitive to tolerance choices.

8 Original Counterfactual: Subsidies $\times$ Borrowing Constraints

8.1 Motivation

The counterfactual examines how investment subsidies—a common policy in the real world—interact with borrowing constraints. The concern, motivated by Table 14, is that early-investment subsidies can be regressive: a marginal subsidy mostly rewards households already making an interior early-investment choice, while the most constrained households, sitting at or near a corner, have less room to take advantage of it. Since these constrained households are largely lower-educated and lower-income, the question I ask is whether relaxing their borrowing constraints, on top of the subsidies, can move them off that corner and help close the educational investment gap.

The design crosses four borrowing regimes (Base; young-parent limit +\$2,500; old-parent limit +\$2,500; all limits relaxed, $\gamma = 0.99$) with three subsidy policies ($s_1 = 0.10$; $s_2 = 0.026$; both), using the same stationary-distribution methodology as Tables 11/12: each regime’s no-subsidy economy is solved to its own stationary distribution, the subsidy is solved on top of the regime,

the short run applies the subsidized rules to the regime's stationary distribution, and the long run re-converges the distribution. All percent changes are measured *within* regime, against that regime's own no-subsidy economy.

8.2 Short-Run Effects

Children of HS dropouts						Children of college grads					
		$\bar{i}_1$	$\bar{i}_2$	Coll	$\bar{W}_3$			$\bar{i}_1$	$\bar{i}_2$	Coll	$\bar{W}_3$
Base	s_1	77.4	29.2	67.8	3.8	Base	s_1	44.6	11.0	19.3	7.4
	s_2	17.0	41.7	59.0	2.7		s_2	6.1	11.1	18.3	2.7
	s_1+s_2	89.7	58.5	103.8	5.7		s_1+s_2	47.1	15.0	25.3	8.4
Young	s_1	91.1	34.9	79.8	4.6	Young	s_1	47.5	12.0	20.9	7.9
	s_2	26.9	48.9	73.8	3.4		s_2	6.5	12.1	20.0	3.0
	s_1+s_2	109.6	68.8	129.2	6.9		s_1+s_2	49.3	15.8	26.7	8.8
Old	s_1	89.3	37.3	86.5	4.6	Old	s_1	43.5	9.1	16.3	6.9
	s_2	27.8	49.0	80.3	3.4		s_2	5.6	10.2	16.7	2.5
	s_1+s_2	98.2	62.5	116.9	6.2		s_1+s_2	46.1	13.1	21.9	7.9
All	s_1	98.4	62.5	135.5	6.3	All	s_1	55.2	13.5	23.2	9.0
	s_2	42.2	114.8	135.5	7.8		s_2	13.6	13.9	23.2	4.2
	s_1+s_2	104.2	128.7	183.7	10.3		s_1+s_2	54.6	15.7	26.4	9.3

Table 23: Short-run effects by parental education. Regimes: Base; Young/Old limit +\$2,500; All relaxed ($\gamma = 0.99$). $s_1 = 0.10$, $s_2 = 0.026$. All values represent a % change relative to the baseline economy with base borrowing limits and no subsidies. Dropout Coll percentages are off small baselines.

8.3 Long-Run Effects

Children of HS dropouts						Children of college grads					
		$\bar{i}_1$	$\bar{i}_2$	Coll	$\bar{W}_3$			$\bar{i}_1$	$\bar{i}_2$	Coll	$\bar{W}_3$
Base	s_1	82.1	31.5	77.5	4.1	Base	s_1	34.9	4.7	8.8	5.1
	s_2	-87.0	-35.7	-87.9	-5.0		s_2	-3.7	3.7	5.7	0.1
	s_1+s_2	-69.4	-30.7	-78.6	-4.1		s_1+s_2	30.2	7.0	11.4	4.9
Young	s_1	79.4	27.7	62.5	3.8	Young	s_1	34.6	4.9	9.0	5.1
	s_2	-87.9	-36.8	-89.3	-5.1		s_2	-4.0	3.9	5.9	0.1
	s_1+s_2	-71.2	-31.4	-79.2	-4.3		s_1+s_2	30.4	7.3	11.9	5.0
Old	s_1	103.1	44.9	102.8	5.5	Old	s_1	35.4	5.1	9.6	5.2
	s_2	-84.7	-33.2	-85.5	-4.7		s_2	-1.9	5.0	7.9	0.5
	s_1+s_2	-68.3	-28.4	-75.3	-4.0		s_1+s_2	31.6	7.3	11.5	5.1
All	s_1	67.0	50.2	104.8	4.7	All	s_1	54.9	13.7	23.2	9.1
	s_2	—	—	—	—		s_2	10.8	12.7	21.0	3.6
	s_1+s_2	—	—	—	—		s_1+s_2	53.4	15.7	25.8	9.2

Table 24: Long-run effects by parental education. Regimes and policies as in Table 23. All values represent a % change relative to the baseline economy with base borrowing limits and no subsidies. Dashes (—): under the fully relaxed regime with a college subsidy, the HS-dropout parent group has zero mass in the long-run steady state, so conditional moments are undefined—the group selects out of existence entirely.

8.4 Findings

Short run. In percent terms the subsidies look progressive when looking at parental investment expenditure, yet when children enter the labor market, the policy is regressive outside of late-investment subsidies where early investment changes are meager for children of college graduates. As we see, even with expansion in education subsidies and fully relaxed borrowing constraints, the inter-generational transmission of ability plays an important role in children’s labor market prospects. Still, the sizable shift in college graduation rates and progressive labor market outcome changes in the s_2 only and $s_1 + s_2$ case shows how subsidies can play an important role in mitigating the income uncertainty channel in the investment gap.

Long-run. In the long-run, children of HS dropouts see a major distributional shift. Thus when looking at subsidies that promote late-investment ($s_2, s_1 + s_2$), the reversal of signs exhibits the distributional shift to those agents with the unlucky ability and wage shocks. However, this shift is not as large in the early investment case, which is why the outcomes across the board are more intuitive. This distributional shift is most notable in the fully relaxed case, where there is no longer any agents within the HS dropout group. In either case, outcomes are most affected by the relaxation of the old’s constraints, and relaxation of all constraints, consistent with the story of older parents hitting their constraints at a higher rate than the young parents.

Takeaway. Under baseline constraints the subsidies are regressive in levels, with college-graduate families capturing roughly $1.5\times$ the labor market gains. Relaxing borrowing constraints

in the $s_2, s_1 + s_2$ case reverses this and concentrates the gains on the previously constrained families.

9 Conclusion

For the majority of tables, the replication was successful up to differences attributable to solution-method noise. The EGM method for solving this model delivered a **$7.5\times$ speedup**, allowing for finer resource and early-investment grids and for SMM calibration. **EGM** was also found to be **better when working with constraints**: optimizer-based tools approach borrowing limits but do not explicitly hit them, unlike the clamping built into EGM. Three tables show non-trivial differences from the published results, for two different reasons. For Table 7, tightening optimizer tolerances and adding explicit corner checks brings the two Julia solvers into near-exact agreement with each other; the remaining gap against the published constrained fractions is attributable to the authors' Nelder–Mead tolerance at the borrowing corners, where their optimizer approaches but never exactly hits the limit. For Table 13 and 15, the two Julia solvers agree to a decimal for nearly every entry. Finally, the methods adopted from the paper's code, the z -state consolidation, per- (h, t) grids, and closed-form log-normal expectations reduced discretization error and improved code readability, and will be nice tricks to be used in my future work.